

Unit-3 Scripting WT

* Javascript: It also known as the scripting language for webpage.

- JS is a programming language: we use it to give instructions to the computer.
- JS can be used for client-side development and as well as server side development

Input $\rightarrow$ computer $\rightarrow$ output
English $\rightarrow$ translator $\rightarrow$ 0 and 1

* Variable:

- In JS, a variable is a container that stores data values.

* Document:

- It's one of the key part of the DOM. The document object provides a way to interact with HTML elements of web page.
- An object that give access to and manipulate the currently loaded HTML document.

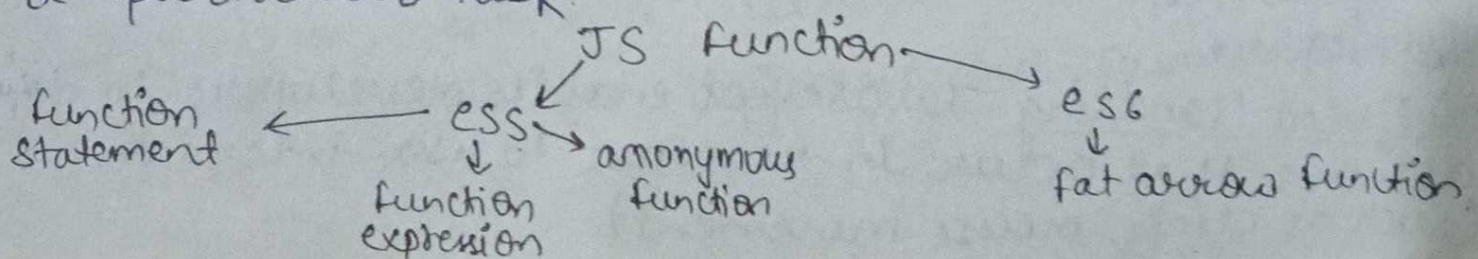
* Statement in JS:

- Javascript statement are composed of values, operator, expressions, keywords and comments

Ex - let x, y, z // Statement 1
 x = 5; // Stat 2
 y = 6; // Stat 3
 z = x + y; // Stat 4

* functions in JS:

- A JS function is a block of code designed to perform a particular task.



Frame work and Libraries

Javascript has a rich ecosystem of frameworks & libraries (such as React, Angular etc)

• function to check checkbox textbox is empty or not:

```
function is TextBoxEmpty(textBox) {  
  if (textBox.value.trim() === ' ') {  
    return true; // text box is empty  
  }  
  else {  
    return false; // textbox is not empty  
  }  
}
```

• Points:-

- `textBox.value` → retrieves the value of the textbox
- `.trim()` → This ensures that an input containing only space or tabs

* objects in Javascript:

- In Javascript, objects are king. If you understand objects, you understand Javascript
- An object is a collection of properties

Syntax of objects in JS:

```
const obj = {  
  // Code  
};
```

* AJAX:

Ques:- Explain the working of AJAX along with its application mention a suitable example.

Ans • AJAX stands for Asynchronous Javascript and XML

- button is clicked).
- (2) An XML HttpRequest object is created by Javascript
 - (3) The XML HttpRequest object sends a request to a web server
 - (4) The server processes the request
 - (5) The response is ready by JS
 - (6) Proper action is performed

• Application of AJAX:

- updating a webpage without reloading the page.
- Requesting data from the server after the page has been loaded.
- Sending data to the servers in the background without disturbing UI or other processes.

Example of AJAX:

- Request XML files
- Retrieve server data with PHP
- Retrieve Database Information
- Date in Javascript:

Date objects encapsulate an integrate number that represent milliseconds since the midnights at the beginning of Jan 1, 1970

Ques 3 - Design a Javascript program to display the current day and times in the following format

Today is Monday

Current time is : 11 Am: 50: 50

sol → `var. today = new Date();`

`var day = today.getDay();`

`var daylist = ["sunday", "monday", "Tuesday", "Wednesday", "Thursday", "Friday", "saturday"];`

`console.log("Today is: " + daylist[day] + ".")`

`var. hour = today.getHours();`

`var minute = today.getMinutes();`

`var second = today.getSeconds();`

Examples

- Math.E // return Euler's number
- Math.PI // return PI value
- Math.SQRT 2 // return square root of 2
- Math.LN 2 // return the natural logarithm of 2
- Creating Date object:

- Date objects are created with the new Date() constructor
- New Date() creates a date object with the current date and time.
- Javascript stores dates as no of milliseconds since Jan 1, 1970
- One day (24 hr) is 86,400.000 milliseconds.

* Internet: Defined as a global network of linked computers, servers, phones and smart application that communicate with each other using the transmission control protocol standard to enable the fast exchange of information and files, along with other type of service

* Internet Addressing:

• IP Address:

- IP address stands for Internet Protocol Address.
- It is an identifying number that is associated with a specific computer or computer network.
- When connected to the internet, the IP address allow the computer to send and receive information.

Types of IP Address:

- (1) IPV4
- (2) IPV6

URL stands for Uniform Resources Locator
It points to a resource on the world wide web. For
example.
<https://www.youtube.com/watch?v=Uj4y73y5d64>

[illegible]

A URL contains many information:

Protocol: In this case http is protocol

- Server name or IP address - www.javapoint.com

- Port number - It is an optional attribute. If we write `http://www.multiautom.com:80/wtfile/`, 80 is the Port number. If Port no is not mentioned in the URL it return -1.

- File Name or directory name - In this case, index.js is the file name

* Datagram -

- * Datagram -
- Datagram are collection of information sent from one device to another device via the established network.

- when the datagram is sent to the targeted device, there is no assurance that it will reach to the target device safely and completely.

- The UDP Protocol is used to implement the datagram in Java